ALICE L WALTON SCHOOL OF MEDICINE

Cancer Immune Evasion

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Adapted by: Chelsey Deel, Stephen Nix

Editors: Jordan Finn, Russell Johnson

Adapted from: Cancer Growth and Metastasis

Listen to this Brick

0:00 / 11:16 1x

Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Explain the mechanisms by which tumor cells evade immune detection.

CASE CONNECTION

A patient presents to the oncology clinic for advanced stage lung cancer. They have tried many rounds of chemotherapy with little success. Today, the oncologist would like to speak to the patient about Keytruda (pembrolizumab), which is an antibody against the PD-1 receptor. What do you tell the patient about this medication and why it may be helpful?

Consider your answers as you read, and we will revisit this case at the end of the brick.

GO TO CONCLUSION ↓

Previous studies have coined the term "immune surveillance", which implies that a normal function of the immune system is to constantly "scan" the body for emerging malignant cells and destroy them. As we navigate the hallmarks of cancer (Figure 1), we have identified that one of the important tenets of cancer is its ability to evade the immune system (avoiding immune destruction).

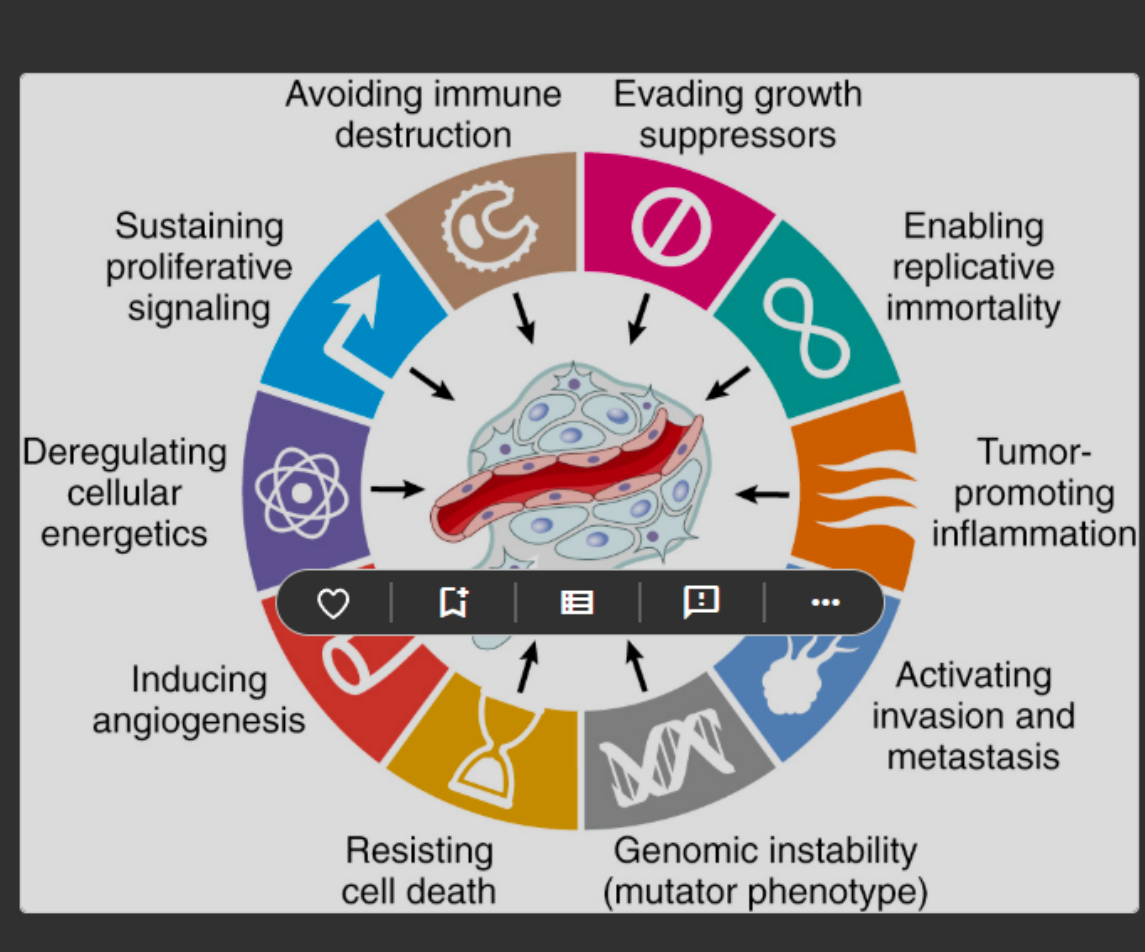
Avoiding immune destruction
Evading growth suppressors
Sustaining proliferative signaling
Enabling replicative immortality
Deregulating cellular energetics
Tumor-promoting inflammation
Inducing angiogenesis
Activating invasion and metastasis
Resisting cell death
Genomic instability (mutator phenotype)

Figure 1. Hallmarks of Cancer

How do cancer cells do this? As it turns out, they have many ideas!

How Does Cancer Evade the Immune System (LO #1)?

Antigen Masking

Antigen masking is one way that tumor cells hide from the immune system. Many tumor cells secrete an excess of extracellular matrix material called glycocalyx. This thick coating on cells "masks" the antigens (molecules) on cancer cells, making it more difficult for immune cells to target them for destruction (Figure 2).

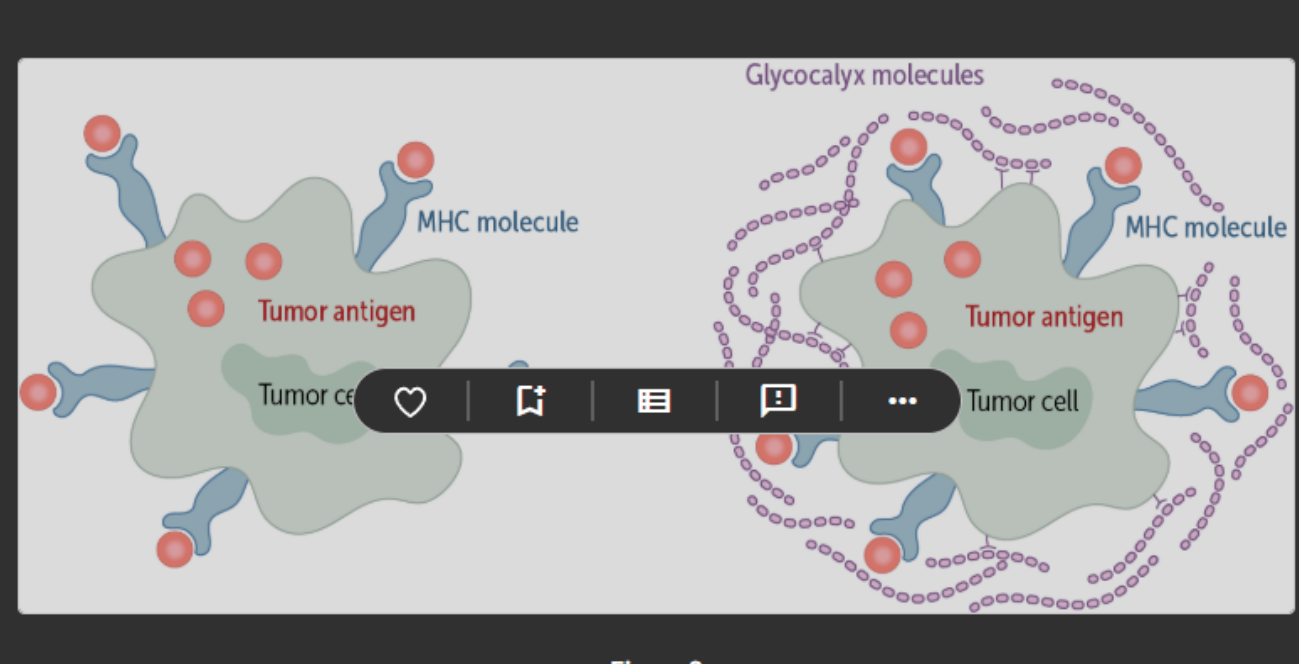
MHC molecule
Tumor antigen
Tumor cell
Glycocalyx molecules
MHC molecule
Tumor antigen
Tumor cell

Figure 2

Outgrowth of Antigen Variants

If a subclone of tumor cells is highly immunogenic, meaning it can easily present as foreign to host immune cells, the subclone group will be eliminated. In contrast, some tumor subclones can lose the expression of surface antigen or downregulate antigen expression. These tumor cells will not express an antigen that can be recognized by any of our immune cells. Although this has been observed in cancers such as melanoma, the exact mechanism is unknown (Figure 3).

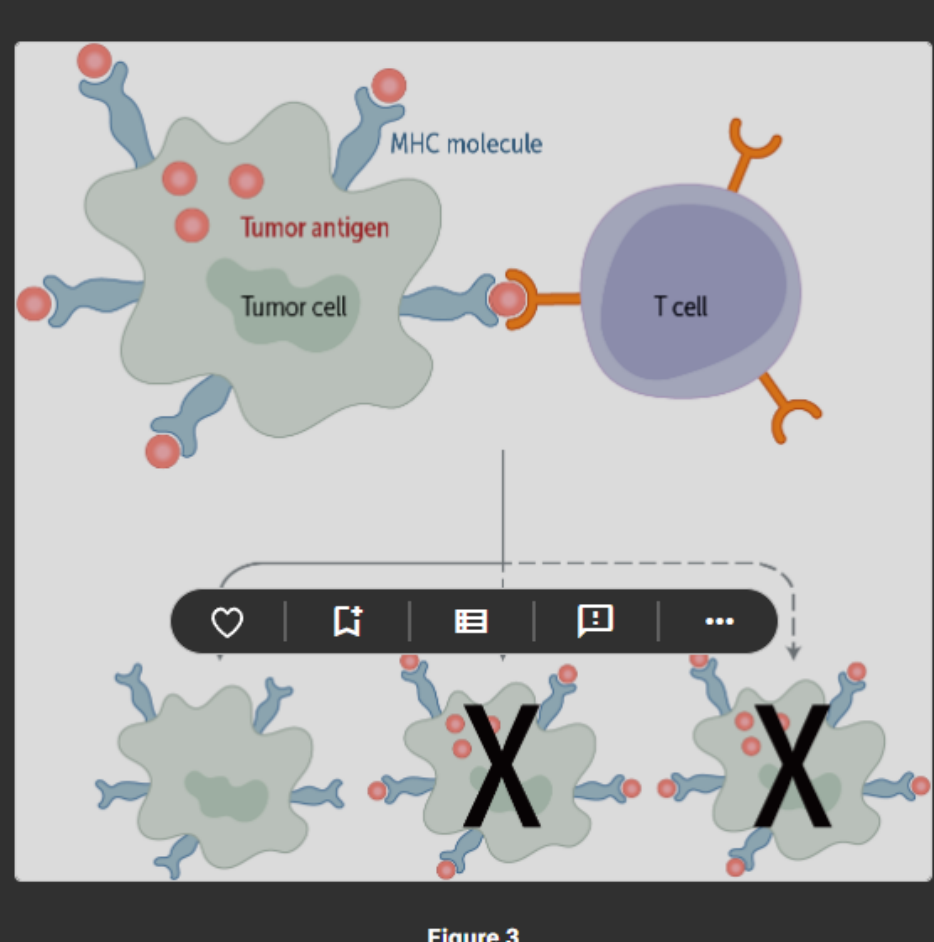
MHC molecule
Tumor antigen
Tumor cell
T cell
X
X

Figure 3

CLINICAL CORRELATION

This phenomenon even extends into cancer immunotherapy. CAR-T (chimeric antigen receptor T-cell) therapy has emerged as a potential treatment for several malignancies including B-cell malignancies, such as diffuse large B-cell lymphoma and B-lymphoblastic leukemia/lymphoma. In this therapy, the patient's own T-cells are engineered to target a specific antigen, like CD19 on B cells. However, a well-known challenge for this therapy is patient cancer cells may stop expressing CD19 on their surface and, thus, there is no target for the T cells.

Loss of MHC Molecules

In some cases, tumor cells can change their identity to avoid immune system defenses. Cytotoxic T cells are white blood cells that fight infection and work to kill foreign cells and cancer cells. These T cells identify target cells by looking for MHC class I molecules. MHC is a receptor present on all cells that displays antigens on the cell surface. Some tumor subclones undergo mutations to lose the MHC molecules so they are no longer recognized by cytotoxic T cells (Figure 4).

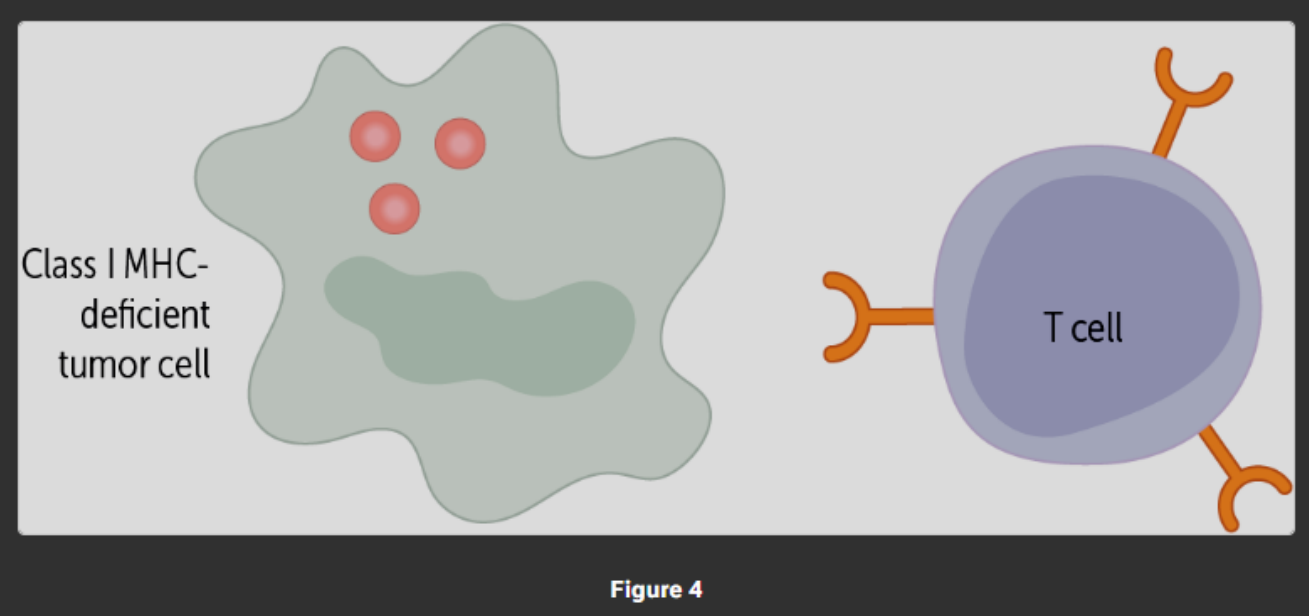
Class I MHC-deficient tumor cell
T cell

Figure 4

Lack of Co-Stimulation

In general, a T cell can only be activated after receiving two signals; this ensures that T cells are not destroying host cells (Figure 5). The first signal is from antigen activation presented by MHC class I or MHC class II molecules. The second signal is a co-stimulatory signal from the antigen-presenting cell (APC).

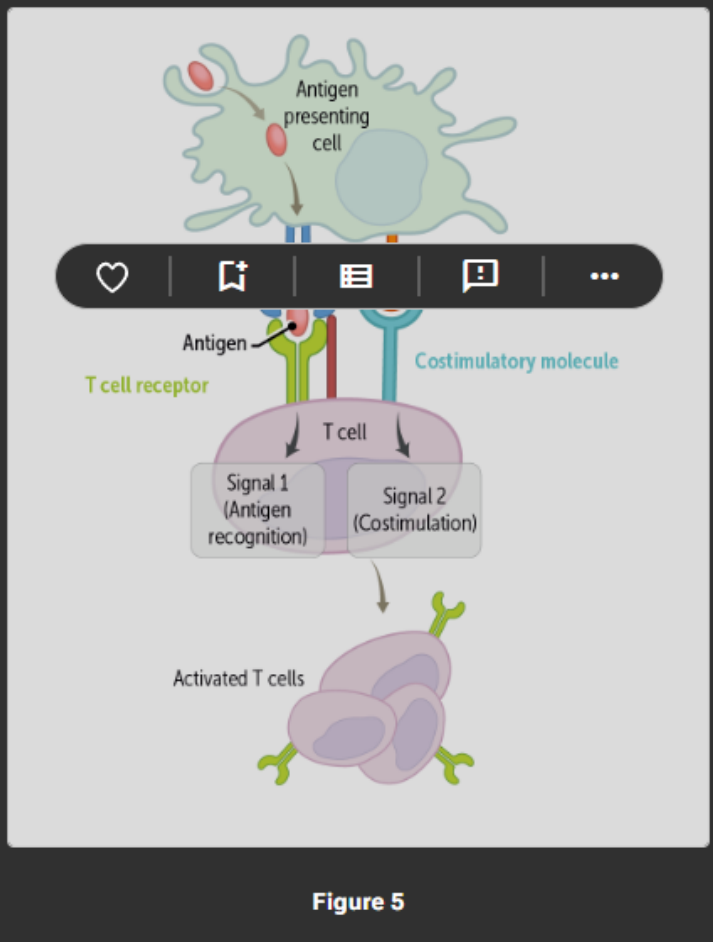
Antigen presenting cell
Antigen
T cell receptor
Costimulatory molecule
T cell
Signal 1 (Antigen recognition)
Signal 2 (Costimulation)
Activated T cells

Figure 5

If this co-stimulatory signal is not received, the T cell will undergo apoptosis, programmed cell death (Figure 6). Tumor cells can rid themselves of this co-stimulatory signal, allowing them to evade the immune system and cause destruction of T cells.

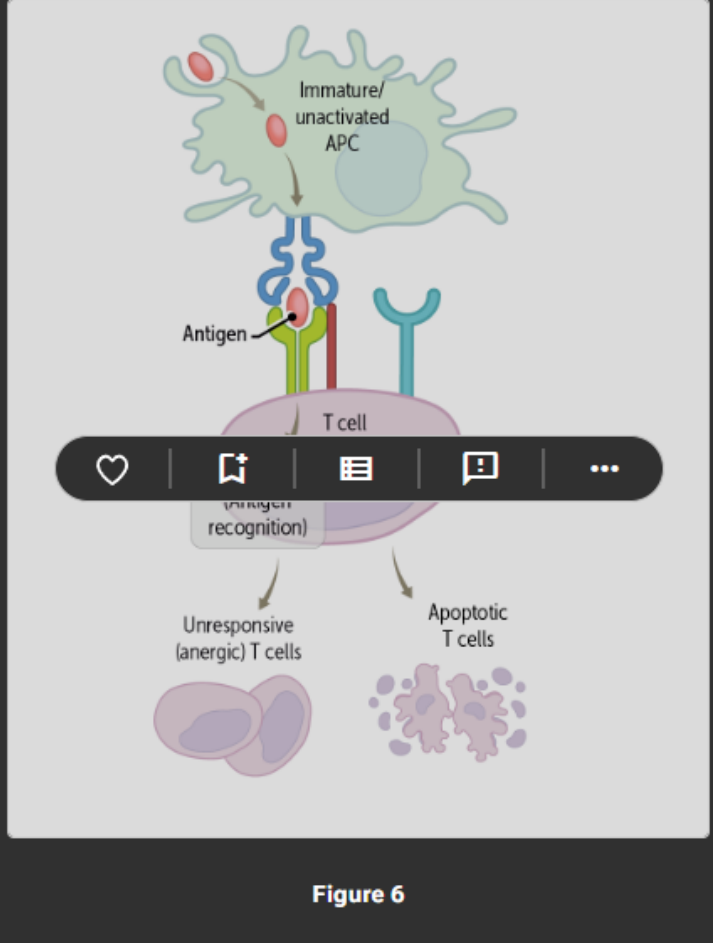
Immature/ unactivated APC
Antigen
T cell
recognition)
Unresponsive (anergic) T cells
Apoptotic T cells

Figure 6

CLINICAL CORRELATION

Cancer cells can upregulate expression of the inhibitory receptor CTLA-4. CTLA-4 binds its ligands, the B7 molecules, from antigen presenting cells and reduces the engagement of the CD28 costimulatory receptor (Figure 7).

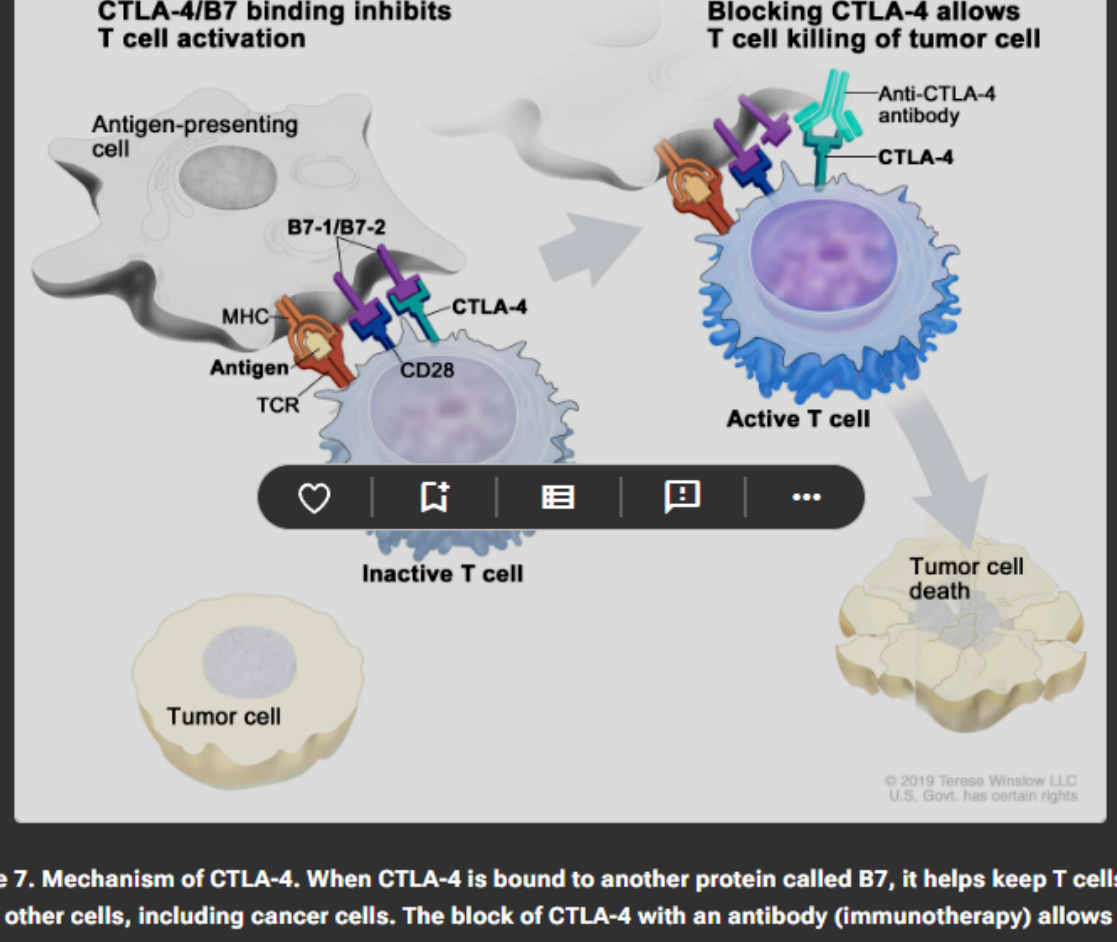
CTLA-4/B7 binding inhibits T cell activation
Antigen-presenting cell
B7-1/B7-2
MHC
CTLA-4
Antigen
CD28
TCR
Inactive T cell
Tumor cell
Blocking CTLA-4 allows T cell killing of tumor cell
Anti-CTLA-4 antibody
CTLA-4
Active T cell
Tumor cell death
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Figure 7. Mechanism of CTLA-4. When CTLA-4 is bound to another protein called B7, it helps keep T cells from killing other cells, including cancer cells. The block of CTLA-4 with an antibody (immunotherapy) allows T cells to kill tumor cells.

Immunosuppression

The immune response is regulated by cytokines, which are small proteins that function as signaling molecules. One of these cytokines is transforming growth factor β (TGF-β), a potent immunosuppressant capable of suppressing the activities of B cells, T cells, dendritic cells, and macrophages. Some tumor cells secrete TGF-β in large quantities, suppressing the immune system (Figure 8).

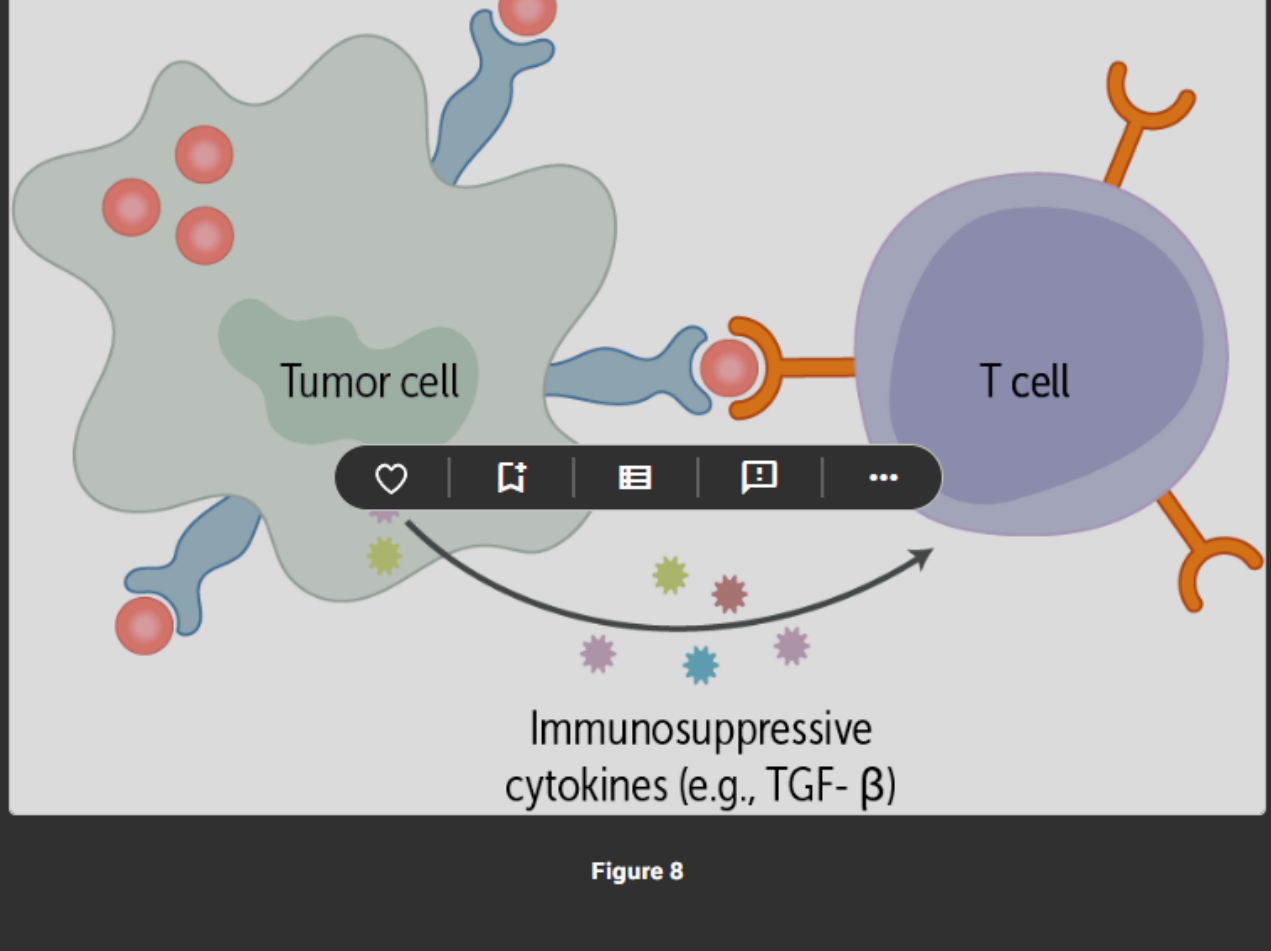
Tumor cell
T cell
Immunosuppressive cytokines (e.g., TGF- β)

Figure 8

What potent immunosuppressant do tumor cells use to evade immune system defenses?

Engagement of Pathways that inhibit T Cell Activation

Tumor cells may inhibit tumor immunity by upregulating negative regulatory "checkpoints" that suppress the immune response.

Cancer cells can upre … D-L2, cell surface proteins that activate the programmed cell death (PD-1) receptor on T cells.

CLINICAL CORRELATION

Knowing these mechanisms, new cancer therapy has emerged, known as "checkpoint blockade", in which antibodies against CTLA-4 and PD-L1 have been used to restore T cell function (Figure 9). This treatment has now been approved for treatment of patients with advanced stage solid tumors and some forms of lymphoma.

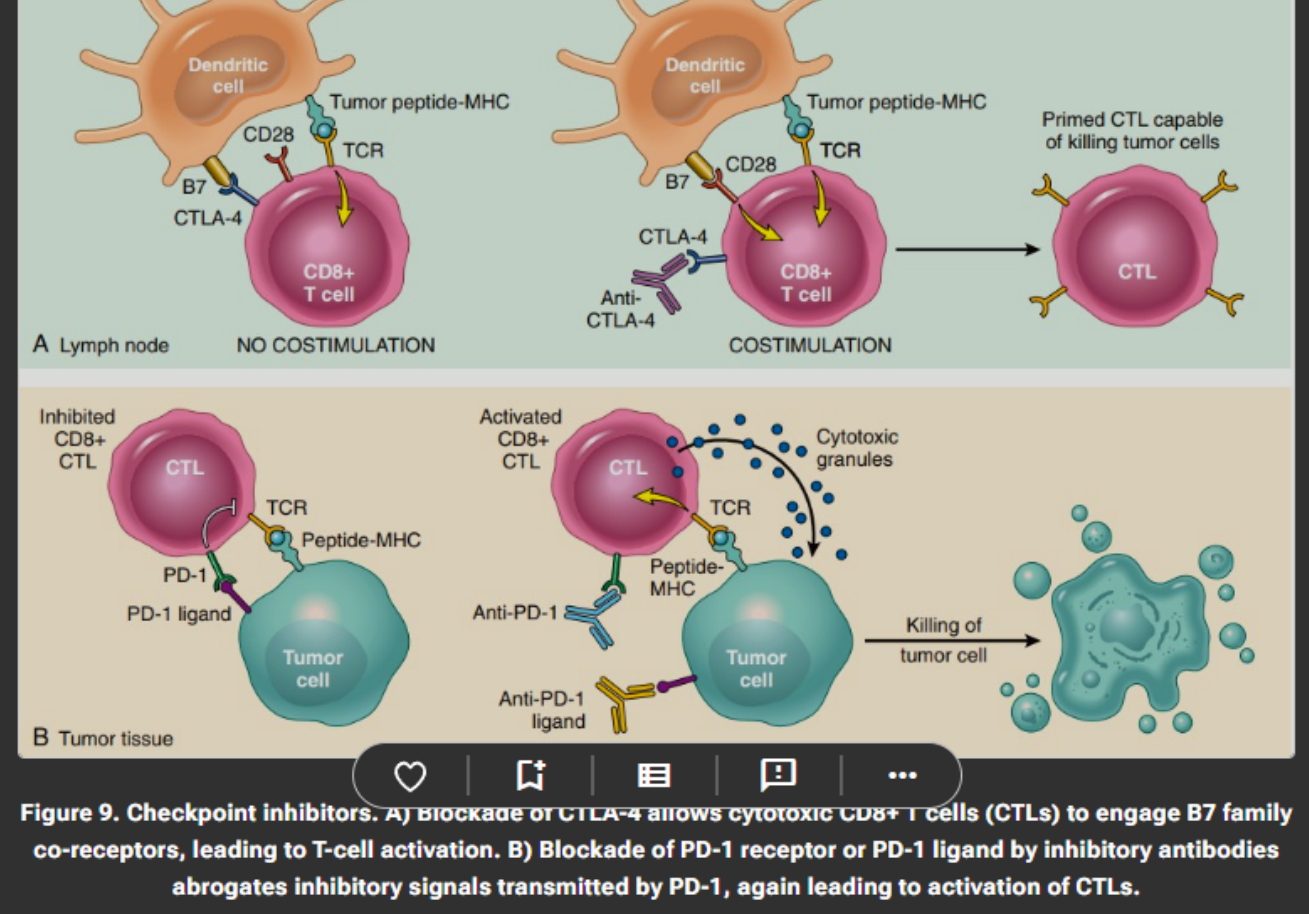
Dendritic cell
Tumor peptide-MHC
CD28
TCR
B7
CTLA-4
CD8+ T cell
A Lymph node
NO COSTIMULATION
Dendritic cell
Tumor peptide-MHC
CD28
TCR
B7
CTLA-4
Anti-CTLA-4
CD8+ T cell
COSTIMULATION
Primed CTL capable of killing tumor cells
CTL
Inhibited CD8+ CTL
CTL
TCR
Peptide-MHC
PD-1
PD-1 ligand
Tumor cell
B Tumor tissue
Activated CD8+ CTL
CTL
Cytotoxic granules
TCR
Peptide-MHC
Anti-PD-1
Tumor cell
Anti-PD-1 ligand
Killing of tumor cell

Figure 9. Checkpoint inhibitors. A) Blockade of CTLA-4 allows cytotoxic CD8+ T cells (CTLs) to engage B7 family co-receptors, leading to T-cell activation. B) Blockade of PD-1 receptor or PD-1 ligand by inhibitory antibodies abrogates inhibitory signals transmitted by PD-1, again leading to activation of CTLs.

Apoptosis of Cytotoxic T Cells

Apoptosis (programmed cell death) is basically a self-destruct button. This process can occur via two different mechanisms: the intrinsic or extrinsic pathway. This concept has been previously reviewed in Cell Death: Apoptosis and Necrosis.

Tumor cells such as melanoma and hepatocellular carcinoma express a receptor that leads to apoptosis via the extrinsic pathway. They express FasL, which binds to the Fas-receptor on T cells, leading to apoptosis of cytotoxic T cells.

CASE CONNECTION

BACK TO INTRODUCTION ↑

You explain to the patient that cancer cells have many ways of trying to get around our body's immune system, including expressing proteins like PD-L1 and PD-L2 that activate the programmed cell death (PD-1) receptor on T-cells. The medication Keytruda is an antibody that binds to the PD-1 receptor on T cells, preventing it from interacting with PD-L1 on cancer cells. Therefore, it gives our immune system a fighting chance to get rid of the cancer cells.

Summary

- The term 'immun … ystem's role in scanning the body for emerging malignant cells and destroying them.
- Cancer cells can evade the immune system through several mechanisms including antigen masking, outgrowth of antigen variants, loss of MHC molecules, lack of co-stimulation, and immunosuppression.
- Some cancer cells place substances on their surface that "mask" the antigens, making it more difficult for immune cells to target them for destruction.
- Cancer cells can lose or downregulate the expression of a surface antigen to avoid immune detection.
- Cancer cells can change their identity by losing MHC molecules, making them unrecognizable to cytotoxic T cells.
- Cancer cells can also inhibit tumor immunity by upregulating negative regulatory 'checkpoints' that suppress the immune response, such as CTLA-4 and PD-L1/PD-L2.
- Some tumor cells secrete the potent immunosuppressant TGF-β in large quantities, suppressing the immune system.
- New cancer therapies, known as 'checkpoint blockade', use antibodies against CTLA-4 and PD-L1 to restore T cell function.

Review Questions

1. A 50-year-old female is diagnosed with stage 4 pancreatic cancer. She has no significant medical history and states she has been healthy throughout her life. She is concerned how her immune system became so weak. You explain to her that the cancer cells were able to undergo mutations that allowed evasion of the immune system. Which of the following is a substance secrete … immune system?

A. ○ FasL

B. ● TGF-β

C. ○ TNF-α

Hide Explanation

The correct answer is TGF-β (B). TGF-β is a potent immunosuppressant that downregulates all lineages of the immune system. Tumor subclones evade the immune system by secretion of this growth factor. FasL (A), or FAS ligand, is a cell-surface protein present on some tumor cells that leads to apoptosis of cytotoxic T cells. TNF-α (C) causes cachexia in malignancy and leads to white blood cell recruitment.

2. Which mechanism allows tumor cells to evade immune detection by masking their antigens?

A. ● Secretion of glycocalyx

B. ○ Loss of MHC molecules

C. ○ Downregulation of antigen expression

D. ○ Apoptosis of cytotoxic T cells

Hide Explanation

The correct answer is secretion of glycocalyx (A). Tumor cells can secrete an excess of extracellular matrix material called glycocalyx, which masks the antigens on cancer cells, making it difficult for immune cells to target them for destruction. Loss of MHC molecules (B) refers to the mutation of tumor cells to lose MHC molecules, making them unrecognizable to cytotoxic T cells. Downregulation of antigen expression (C) is a mechanism where tumor cells lose or reduce the expression of surface antigens, thus escaping immune detection. Apoptosis of cytotoxic T cells (D) is a process where tumor cells cause the programmed cell death of T cells, thereby evading the immune system. None of these other options involve masking of antigens.

How would you rate this Brick?

References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease*. 10th ed. Elsevier; 2021.
- Pastorczak A, Domka K, Fidyt K, Poprzeczko M, Firczuk M. Mechanisms of Imm… Leukemia. *Cancers*. 2021; 13(7):1536.

How would you rate this Brick?

References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease*. 10th ed. Elsevier; 2021.
- Pastorczak A, Domka K, Fidyt K, Poprzeczko M, Firczuk M. Mechanisms of Immune Evasion in Acute Lymphoblastic Leukemia. *Cancers*. 2021; 13(7):1536. https://doi.org/10.3390/cancers13071536

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanisms by which tumor cells evade immune detection.~~

Objective Confidence: 1/1